

Notes for ‘Applications of the Chain Rule’

Examples:

1. We apply the methods of curve sketching to try to understand the Gaussian curve

$$y = e^{-x^2}.$$

The y -intercept is $y(0) = e^0 = 1$. There are no x -intercepts, as all powers of e are nonzero. Observe that

$$\lim_{x \rightarrow \infty} e^{-x^2} = \lim_{x \rightarrow -\infty} e^{-x^2} = \lim_{u \rightarrow -\infty} e^u = 0,$$

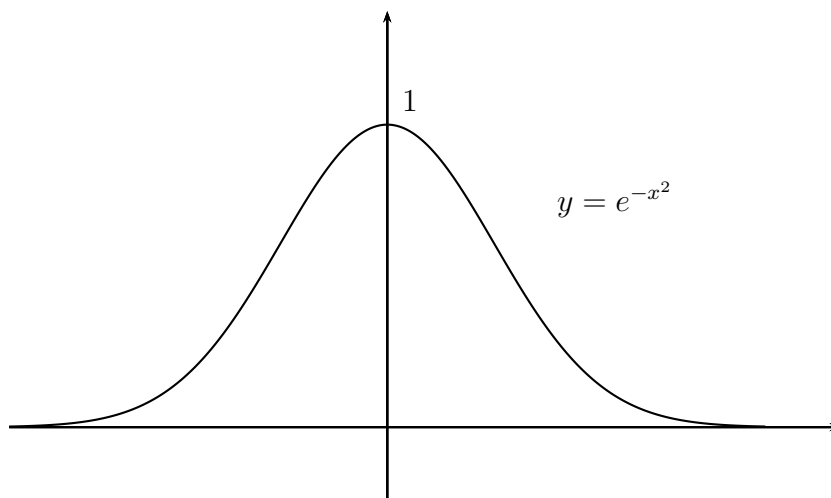
by putting $u = -x^2$, and observing that $u \rightarrow -\infty$ as $x \rightarrow \infty$ and as $x \rightarrow -\infty$. Hence the x -axis is a horizontal asymptote. Observe also that $\frac{du}{dx} = -2x$, so, by the Chain Rule,

$$y' = \frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \frac{d}{du}(e^u) \frac{du}{dx} = e^u(-2x) = -2xe^{-x^2}.$$

Hence $y' = 0$ precisely when $x = 0$, since the factor e^{-x^2} is always positive. For nonzero x , the sign of y' is the opposite of the sign of x , and we get the following sign diagram:

x	0
y'	$+$ 0 $-$

Hence there is a turning point when $x = 0$, so the function associated with the curve achieves a global maximum of $y(0) = 1$. Combining this fact with the asymptotic behaviour, we expect the curve to look something like the following:



We can't be absolutely sure about the concave behaviour until we analyse the second derivative y'' , but we will do that soon, after introducing the Product Rule.

2. Suppose an ornamental ice sculpture in the shape of a cube has been created by freezing 100 litres of water. It has been suspended from the ceiling in a warm room where the temperature has been kept constant. After 3 hours, 10 litres of water have been collected in a tray beneath the ice cube, so that 90% of the ice cube remains. The problem is to estimate the number of hours it takes for the ice cube to melt away completely.

We may use the fact that the rate of melting of the volume of the cube is proportional to the surface area.

Solution: Denote the side-length of the cube by $x = x(t)$ units, which is a function of time t hours, so that the volume V and surface area A are given by the formulae

$$V = V(t) = x^3 \quad \text{and} \quad A = A(t) = 6x^2 ,$$

noting that there are six faces to a cube. We want the time t at which all of x , V and A become zero. We have that

$$\frac{dV}{dt} = kA = 6kx^2 ,$$

for some constant k . Observe that $\frac{dV}{dx} = 3x^2$, so, by the Chain Rule,

$$6kx^2 = \frac{dV}{dt} = \frac{dV}{dx} \frac{dx}{dt} = 3x^2 \frac{dx}{dt} ,$$

so that

$$\frac{dx}{dt} = \frac{6kx^2}{3x^2} = 2k .$$

If the derivative of a function is constant, then the slopes of all tangent lines at all points are constant, so it is clear (intuitively), that the graph of the function must become a straight line. Applying this observation here, we have

$$x = 2kt + C$$

for some constant C . (This step that we have appealed to intuitively, is part of a general phenomenon, known as *integration*, which is explored in detail in the final module of this course.)

The cube loses 10% of its volume after 3 hours, so we have $V(3) = \frac{9}{10}V(0)$. Hence

$$(6k + C)^3 = V(3) = 0.9V(0) = 0.9(x(0))^3 = 0.9(2k(0) + C)^3 = 0.9C^3 ,$$

so that $6k + C = \sqrt[3]{0.9} C$, and it follows, by rearranging and solving for C , that

$$C = \frac{6k}{\sqrt[3]{0.9} - 1} .$$

We want t such that $x(t) = 0$, so we get

$$0 = 2kt + C = 2kt + \frac{6k}{\sqrt[3]{0.9} - 1} = k \left(2t + \frac{6}{\sqrt[3]{0.9} - 1} \right) .$$

Cancelling k (some nonzero constant), and rearranging, we finally get

$$t = \frac{3}{1 - \sqrt[3]{0.9}} \approx 86.9 .$$

Hence we expect it takes nearly 87 hours for the ice cube to melt away completely.

(There is a hidden subtlety: the argument that derived the linear formula for x as a function of t used implicitly the fact that x is nonzero. However a linear function is continuous, so we can use the same formula for x even when x becomes zero.)